

Ядро после запуска ищет программу init. Она хранится в корневой файловой системе.

В лабораторной работе надо написать свой init процесс.

1) Реализация процесса init

```
grogu@osio:~/project/test$ file init
init: ELF 32-bit LSB executable, ARM, EABI5 version 1 (GNU/Linux), statically linked, BuildID[sha1
981b85152eb6bc29a8cb, for GNU/Linux 3.2.0, not stripped
grogu@osio:~/project/test$
```

```
#include <stdio.h>
#include <unistd.h>
```

```
int main(){
    printf("Test init!!!\n");
    sleep(20);
}
```

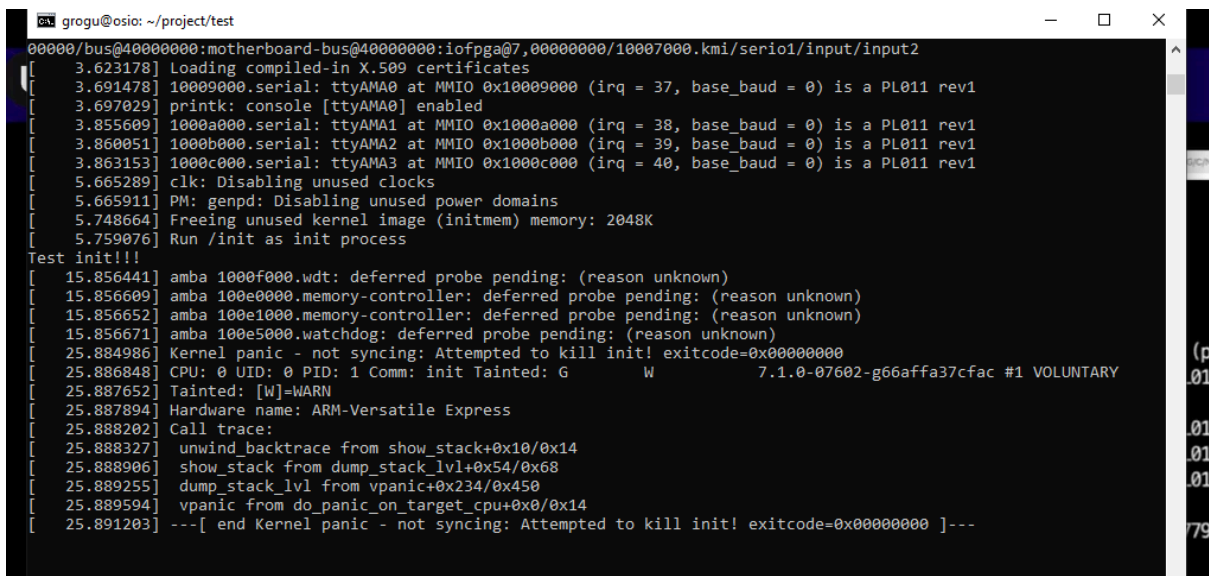
2) создание минимального initramfs (начальный RAM-диск), который содержит только один файл — исполняемый файл.

```
echo init | cpio -o -H newc | gzip > initramfs.cpio.gz
```

3) Запуск эмулятора

```
QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -initrd
initramfs.cpio.gz -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0" -nographic
```

4) Выполнение бинарника. Вывело Test init!!! и ядро снова упало.



```
grogu@osio: ~/project/test
00000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10007000.kmi/serio1/input/input2
[ 3.623178] Loading compiled-in X.509 certificates
[ 3.691478] 10009000.serial: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud = 0) is a PL011 rev1
[ 3.697029] printk: console [ttyAMA0] enabled
[ 3.855609] 1000a000.serial: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud = 0) is a PL011 rev1
[ 3.860051] 1000b000.serial: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud = 0) is a PL011 rev1
[ 3.863153] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a PL011 rev1
[ 5.665289] clk: Disabling unused clocks
[ 5.665911] PM: genpd: Disabling unused power domains
[ 5.748664] Freeing unused kernel image (initmem) memory: 2048K
[ 5.759076] Run /init as init process
Test init!!!
[ 15.856441] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 15.856609] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
[ 15.856652] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
[ 15.856671] amba 100e5000.watchdog: deferred probe pending: (reason unknown)
[ 25.884986] Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000
[ 25.886848] CPU: 0 UID: 0 PID: 1 Comm: init Tainted: G      W       7.1.0-07602-g66affa37cfac #1 VOLUNTARY
[ 25.887652] Tainted: [W]=WARN
[ 25.887894] Hardware name: ARM-Versatile Express
[ 25.888202] Call trace:
[ 25.888327] unwind_backtrace from show_stack+0x10/0x14
[ 25.888906] show_stack from dump_stack_lvl+0x54/0x68
[ 25.889255] dump_stack_lvl from vpanic+0x234/0x450
[ 25.889594] vpanic from do_panic_on_target_cpu+0x0/0x14
[ 25.891203] ---[ end Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000 ]---
```

5) Дальше нужно поставить в эмулятор полноценную корневую файловую систему busybox.

```
wget https://busybox.net/downloads/busybox-1.36.1.tar.bz2
```

```
tar -xjf busybox-1.36.1.tar.bz2
```

6) Сборка дефолтного конфига под arm

```
ARCH=arm make defconfig
```

7) По умолчанию busybox собирается с динамическими библиотеками. Нужно поправить конфиг, чтобы он собирался статически.

8) Собираем корневую файловую систему

```
make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- -j$(nproc)
```

```
grogue@osio:~/project/busybox/busybox-1.35.0$ ls
AUTHORS      NOFORK_NOEXEC.sh  busybox_unstripped  editors      mailutils      scripts
Config.in    README            busybox_unstripped.map  examples     make_single_applets.sh  selinux
INSTALL      TODO              busybox_unstripped.out  findutils    miscutils      shell
LICENSE      TODO_unicode     configs              include       modutils       size_single_applets.
Makefile     applets           console-tools        init          networking      syslogd
Makefile.custom  applets_sh       coreutils            klibc-utils  printutils      testsuite
Makefile.flags  arch              debianutils          libbb         procps          util-linux
Makefile.help  archival          docs                  libpwdgrp    qemu_multiarch_testing
NOFORK_NOEXEC.lst  busybox          e2fsprogs            loginutils    runit
grogue@osio:~/project/busybox/busybox-1.35.0$ file busybox
busybox: ELF 32-bit LSB executable, ARM, EABI5 version 1 (GNU/Linux), statically linked, BuildID[sha1]=d0670e0cbe50115f905147dab521c5d1ce6806, for GNU/Linux 3.2.0, stripped
grogue@osio:~/project/busybox/busybox-1.35.0$
```

9) Устанавливаем

```
make install
```

grogu@osio: ~/project/busybox/busybox-1.35.0

```
./_install//usr/sbin/powertop -> ../../bin/busybox
./_install//usr/sbin/rdate -> ../../bin/busybox
./_install//usr/sbin/rdev -> ../../bin/busybox
./_install//usr/sbin/readahead -> ../../bin/busybox
./_install//usr/sbin/readprofile -> ../../bin/busybox
./_install//usr/sbin/remove-shell -> ../../bin/busybox
./_install//usr/sbin/rtcwake -> ../../bin/busybox
./_install//usr/sbin/sendmail -> ../../bin/busybox
./_install//usr/sbin/setfont -> ../../bin/busybox
./_install//usr/sbin/setlogcons -> ../../bin/busybox
./_install//usr/sbin/svlogd -> ../../bin/busybox
./_install//usr/sbin/telnetd -> ../../bin/busybox
./_install//usr/sbin/tftpd -> ../../bin/busybox
./_install//usr/sbin/ubiattach -> ../../bin/busybox
./_install//usr/sbin/ubidetach -> ../../bin/busybox
./_install//usr/sbin/ubimkvol -> ../../bin/busybox
./_install//usr/sbin/ubirename -> ../../bin/busybox
./_install//usr/sbin/ubirmvol -> ../../bin/busybox
./_install//usr/sbin/ubirsvol -> ../../bin/busybox
./_install//usr/sbin/ubiupdatevol -> ../../bin/busybox
./_install//usr/sbin/udhcpd -> ../../bin/busybox
```

You will probably need to make your busybox binary
setuid root to ensure all configured applets will
work properly.

grogu@osio:~/project/busybox/busybox-1.35.0\$

10) Создаем архив

```
find . | cpio -o -H newc | gzip > initramfs.cpio.gz
```

11) Запускаем эмулятор

12) `QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -initrd initramfs.cpio.gz -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0 rdinit=/bin/ash" -nographic`

```
[ 1.976900] mci-pl18x 10005000.mci: mmc0: PL181 manf 41 rev0 at 0x10005000 irq 33,34 (pio)
[ 2.018754] 10009000.uart: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud = 0) is a PL011 rev1
[ 2.059038] printk: console [ttyAMA0] enabled
[ 2.060796] 1000a000.uart: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud = 0) is a PL011 rev1
[ 2.062090] 1000b000.uart: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud = 0) is a PL011 rev1
[ 2.063352] 1000c000.uart: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a PL011 rev1
[ 2.067193] rtc-pl031 10017000.rtc: registered as rtc0
[ 2.068034] rtc-pl031 10017000.rtc: setting system clock to 2026-05-28T13:14:05 UTC (1779974045)
[ 2.075248] uart-pl011 10009000.uart: no DMA platform data
[ 2.103626] Freeing unused kernel memory: 2048K
[ 2.105992] Run /bin/ash as init process
[ 2.121146] input: AT Raw Set 2 keyboard as /devices/platform/bus@4000000/bus@4000000:motherboard/bus@4000000:motherboard:ioppga@7,00000000/10006000.kmi/serio0/input/input0
/bin/ash: can't access tty; job control turned off
~ # [ 2.764170] input: ImExPS/2 Generic Explorer Mouse as /devices/platform/bus@4000000/bus@4000000:motherboard/bus@4000000:motherboard:ioppga@7,00000000/10007000.kmi/serio1/input/input2
~ #
~ #
~ #
~ #
```